

Alan Roybal

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EDUCATION

University of Texas at Dallas

Bachelor of Science in Computer Science

Expected Graduation: December 2026

GPA: 3.7/4.00

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Machine Learning, Computer Architecture, Computer Networks, Artificial Intelligence, Automata Theory, Advanced Algorithms, Compilers

EXPERIENCE

NorthMark Compute and Cloud

Artificial Intelligence Engineering Intern

June 2026 – Present

Dallas, TX

- Architected a centralized **MCP** gateway brokering secure LLM access across **12+** microservices, cutting latency **40%**.
- Benchmarked **on-prem SOTA** models on **GPU** clusters, shaping an AI roadmap projected to save **\$250K** annually.
- Shipped an **agent evals** harness replaying **30+** task suites to score tool use, grounding, and regressions per release.

Townes & Co.

Founding Engineer

April 2026 – Present

Dallas, TX

- Engineered a multi-tenant AI SaaS platform ingesting **5M+** enterprise records/day at **sub-200ms** p99 query latency.
- Provisioned a cloud-native **Kubernetes** mesh autoscaling to **50+** pods, sustaining **99.9%** uptime for lead customer.

General Motors Financial

Software Engineering Intern

May 2025 – May 2026

Arlington, TX

- Engineered a **C#/NUnit/Selenium** suite on **Azure DevOps** automating **50+** endpoints, saving **10** QA hours/week.
- Built an MCP server delivering **sub-50ms** tool routing, stateful session management, and skill injection into Copilot.
- Briefed the **VP of Engineering** on AI enablement, doubling org-wide **GitHub Copilot** adoption from **7** to **15** teams.
- Integrated a **SOAP-to-REST** layer across **10** endpoints, driving a firm-wide shift from manual QA to automated CI.
- Promoted to lead a **7-person** squad; instituted standups and office hours, lifting velocity **25%** and cycle time **30%**.

PROJECTS

CoTTY | Zig, Swift, SwiftUI, Metal, CRDT, TCP, PTY

- Engineered a **Zig** core (PTY, **TCP**, **RGA CRDT**) bridged to SwiftUI via a hand-written C ABI at sub-ms latency.
- Implemented an **ANSI/VT** escape parser and cell grid for pixel-accurate multi-user emulation at **256-color** depth.
- Built a **Metal** GPU pipeline rasterizing the grid at **60fps+** with real-time multi-user cursors, halving render time.
- Designed an **RGA CRDT** from scratch for conflict-free editing among **4+** NAT-traversed peers at **99%** convergence.

Lyria Studio | Electron, React, TypeScript, Vite, Zustand, Google GenAI

- Built an AI music DAW (**Electron**, **React**, **TypeScript**) streaming a node graph into **Google Lyria Realtime**.
- Shipped a live mode triggering **Lyria** in **sub-300ms** from the node graph, sustaining gapless **48kHz** PCM playback.
- Orchestrated a real-time pipeline into a **16-track** timeline with per-track automation and 24-bit/96kHz WAV export.

Earth2Echo | Python, FastAPI, React, WebSockets, Google Gemini API, Multimodal AI, Generative Audio

- Architected an agent execution layer orchestrating tool integration, model state, and **Lyria 2** audio on Gemini APIs.
- Selected for the Google **Gemini Developer Spotlight**; contributions shipped in the **Lyria 3** audio model release.
- Optimized a WebSocket pipeline syncing multimodal media, cutting end-to-end inference latency **50%** to **sub-400ms**.

LEADERSHIP

Director of HackUTD

Largest 24-hr Hackathon in North America | Association for Computing Machinery Dallas

January 2024 – Present

Dallas, TX

- Owned all vendor, sponsor, and engineering decisions for a **1,000+** participant hackathon on a six-figure budget.
- Directed a **40+** person cross-functional team across tech, logistics, finance, and marketing, running **12+** workshops.
- Deployed the **Jury** judging platform to **10** organizations nationwide, owning onboarding at **100%** event-day uptime.
- Drove open-source contributions to the **HackUTD** repo and led **8** developer tooling workshops for **500+** engineers.

SKILLS

Languages: Python, Go, Zig, JavaScript, TypeScript, Rust, Ruby, R, Swift, C++, C#, Java, SQL, HTML, CSS

Frameworks & Libraries: React.js, Next.js, Node.js, Flask, FastAPI, REST APIs, WebSockets, JUnit, Django, Spring Boot

Tools: AWS, Azure, GCP, Claude Code, Codex, Devin, Docker, Git, PostgreSQL, Firebase, JIRA, Kubernetes, CUDA

Certifications: AWS ML Specialist, AWS Solutions Architect Associate, NVIDIA GenAI and LLM Engineering Associate

Concepts: AI/ML Engineering, Distributed Systems, Backend Development, Frontend Development, Fullstack Development, Software Engineering, Networking, Microservices, Infrastructure Engineering, DevOps, Automation, Agile